Decoding Data Science Salary Determinants

Analysis of the Kaggle dataset: Global Data Science Compensation (2020–2023)
by Amab Chaki

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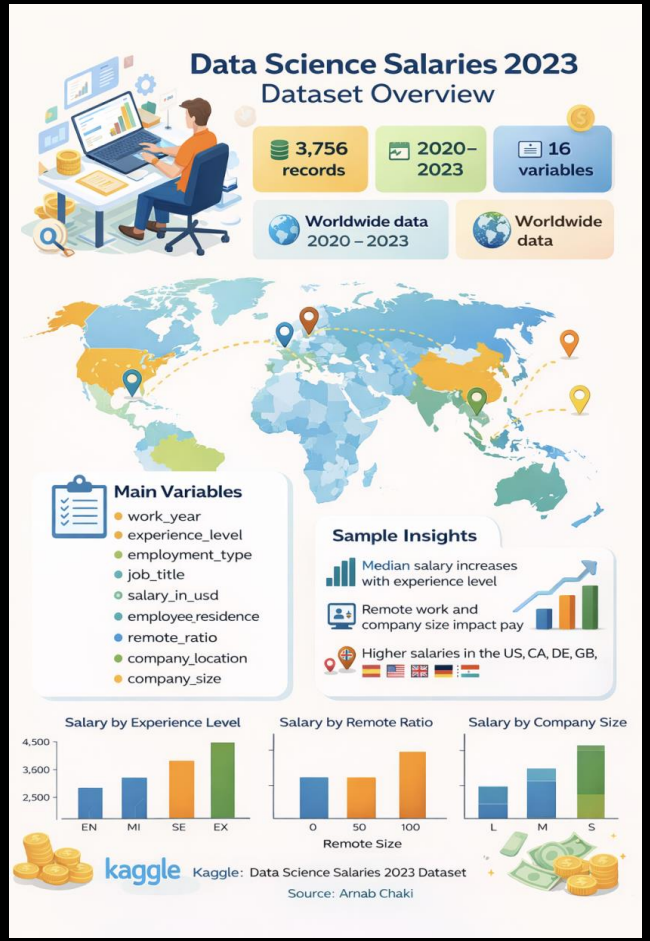
Group 14

Explanation of Dataset

- DS salary records
- **3,756 records, 2020–2023**, multidimensional
- Facilitates analysis of pay by experience, job title, remote work, company size, and geography
- Salary_in_usd supports cross-country comparison

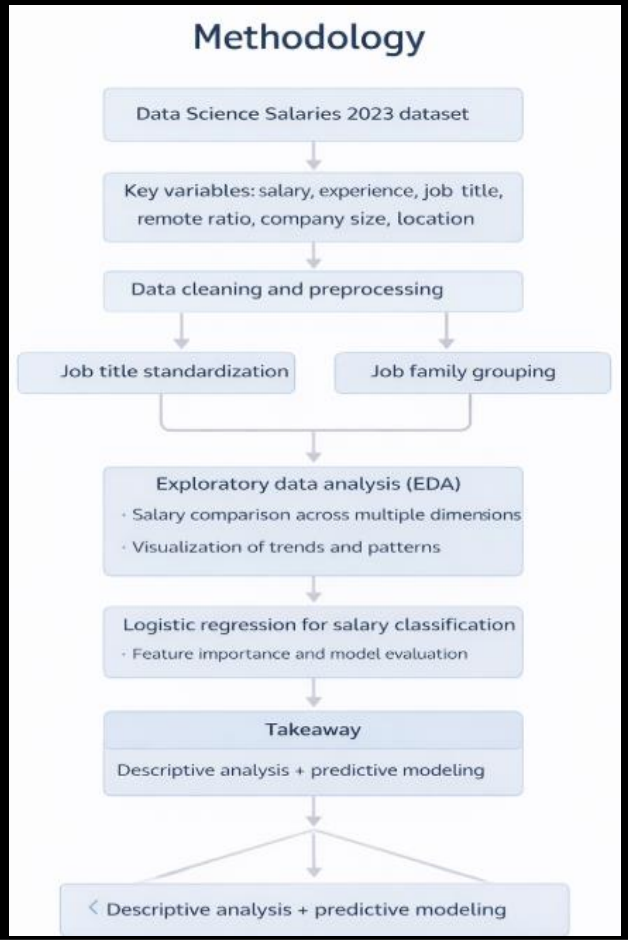


Motivation

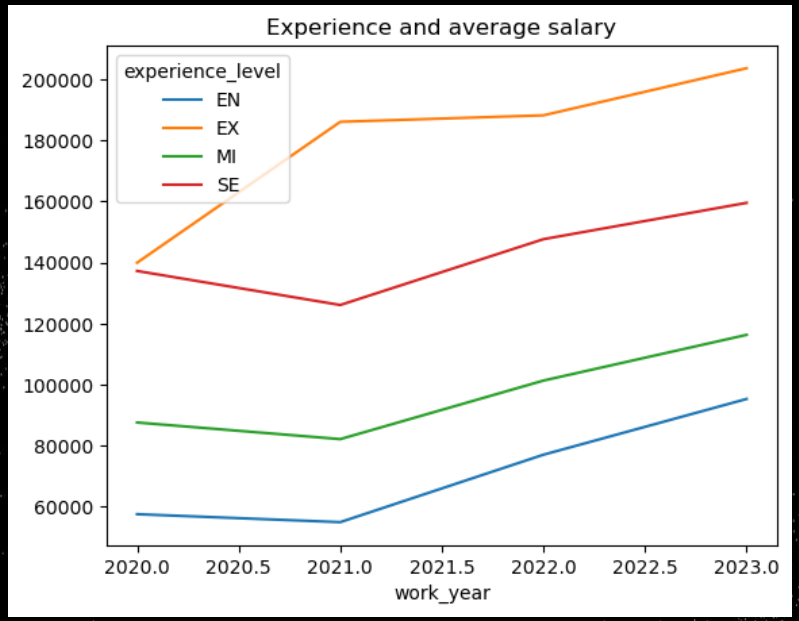
- Salary variance across multiple factors
- Identifying core compensation factors
- Study patterns from various professional perspectives.

Objective

- Analyze salary shifts across experience levels
- Use logistic regression to transition from descriptive insights to salary prediction
- Combining data exploration with statistical modeling to uncover market drivers.



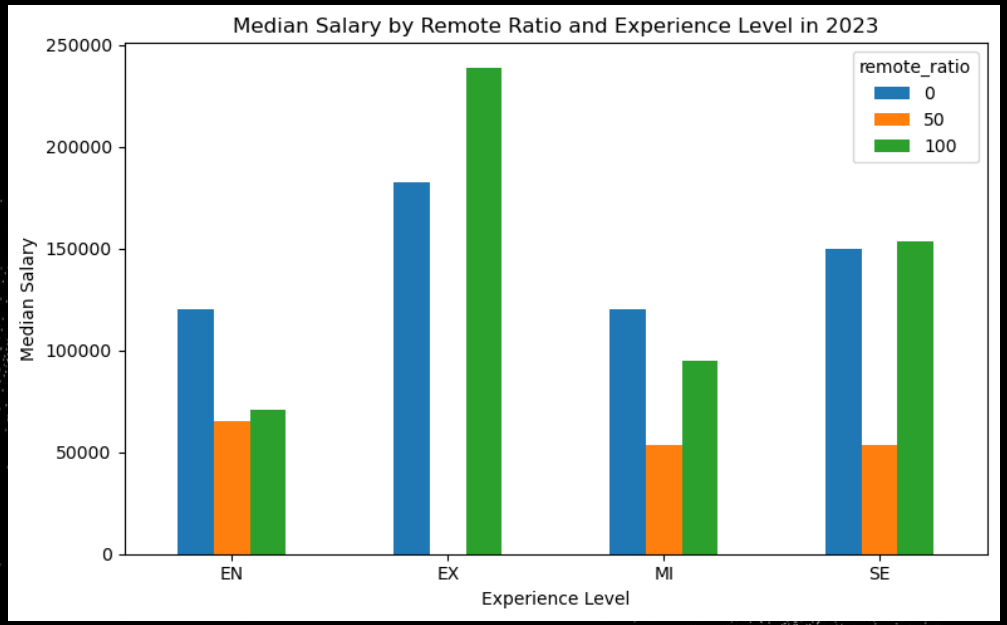
Question 1: How does salary change by experience level across years?



- Increases with experience
- Executives > Entry Level
- Only non-EX dipped in 2021
- Clear gap across levels

Takeaway: Experience level is strongly associated with higher salary

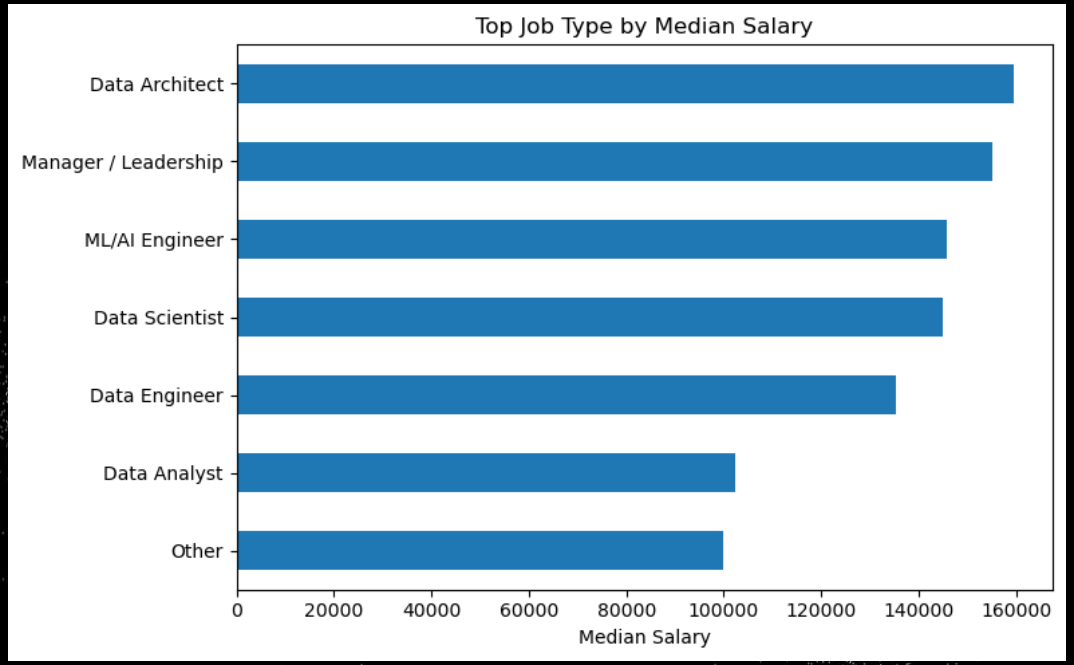
Question 2: Is there a remote work pay premium & does it depend on seniority?



- Remote-work effect differs across seniority levels.
- EX rewarded for remote vs EN punished
- EN and MI show no remote-work premium.
- Hybrid roles generally have the lowest median salary

Takeaway: There is no uniform remote-work pay premium; its impact depends on seniority

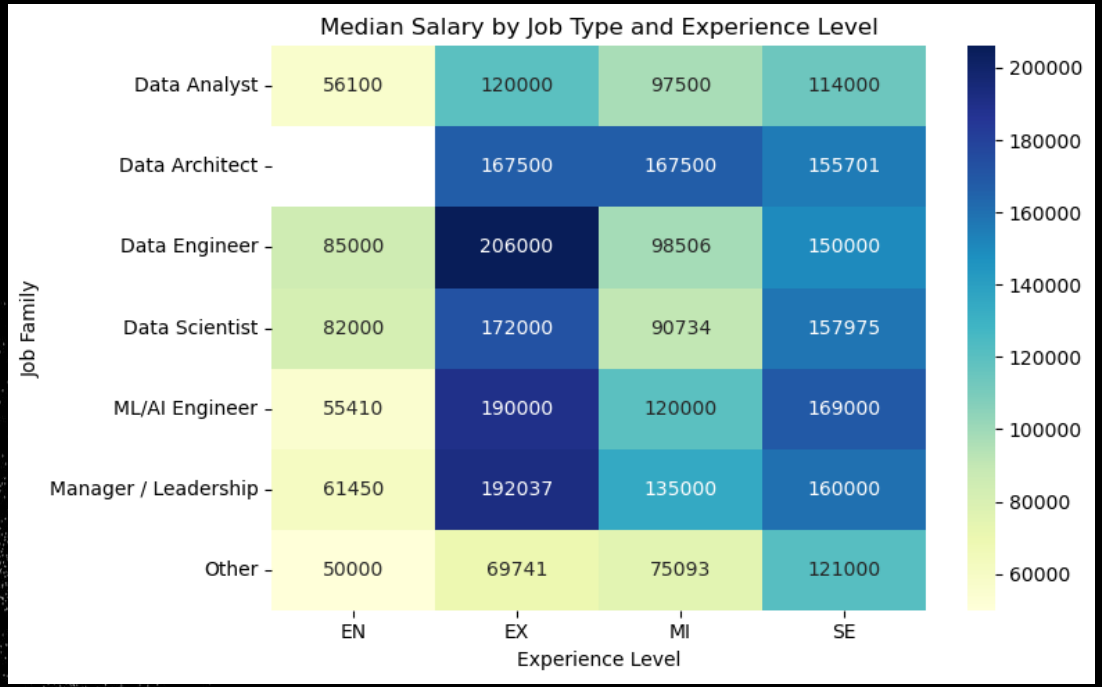
Question 3: Which job titles (or job families) pay the most, and how stable is that ranking?



- **Data Architect has the highest median salary**
- **Manager / Leadership and ML/AI Engineer also rank near the top.**
- **Data Scientist and Data Engineer form the middle tier.**
- **Data Analyst and Other have the lowest median salaries**

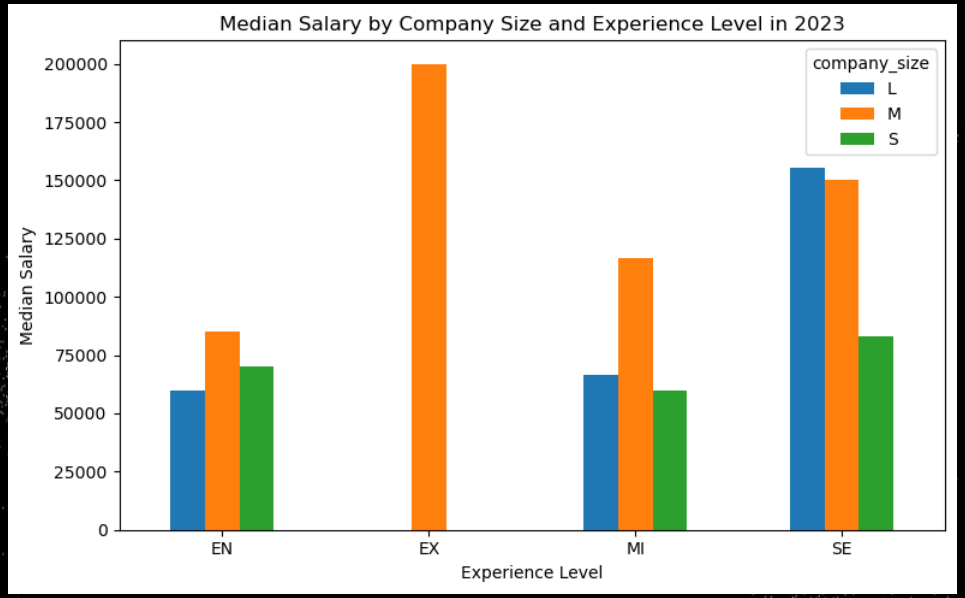
Question 3: Which job titles (or job families) pay the most, and how stable is that ranking?

- **Data Engineer leads at EX**
- **Manager / Leadership and ML/AI Engineer remain strong at SE**
- **Data Analyst relatively lower across most levels**
- **Some job families show greater pay variation by seniority than others**



Takeaway: Job-family salary rankings are not fully stable; seniority changes which roles pay the most

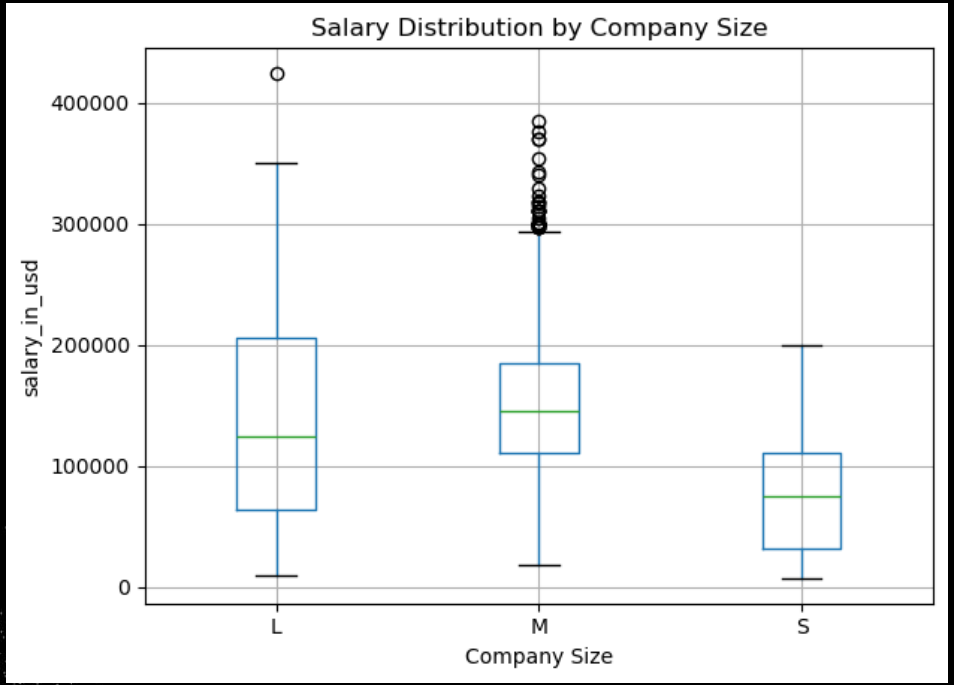
Question 4: Does company size matter after controlling for experience and role?



- **Company size is associated with salary differences**
- **Medium-sized companies often show the highest median salary.**
- **Small companies generally pay less across experience levels.**

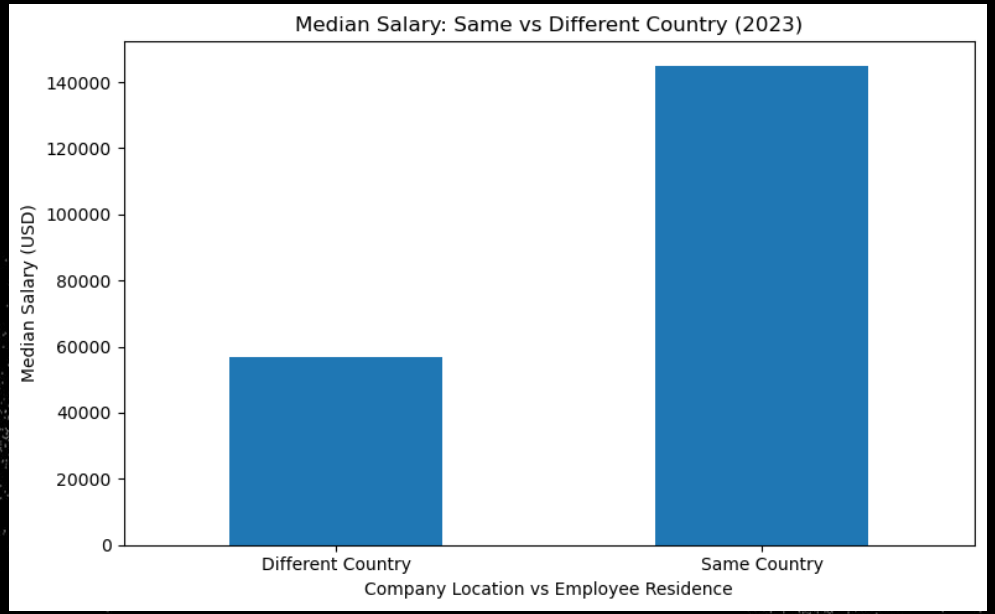
Question 4: Does company size matter after controlling for experience and role?

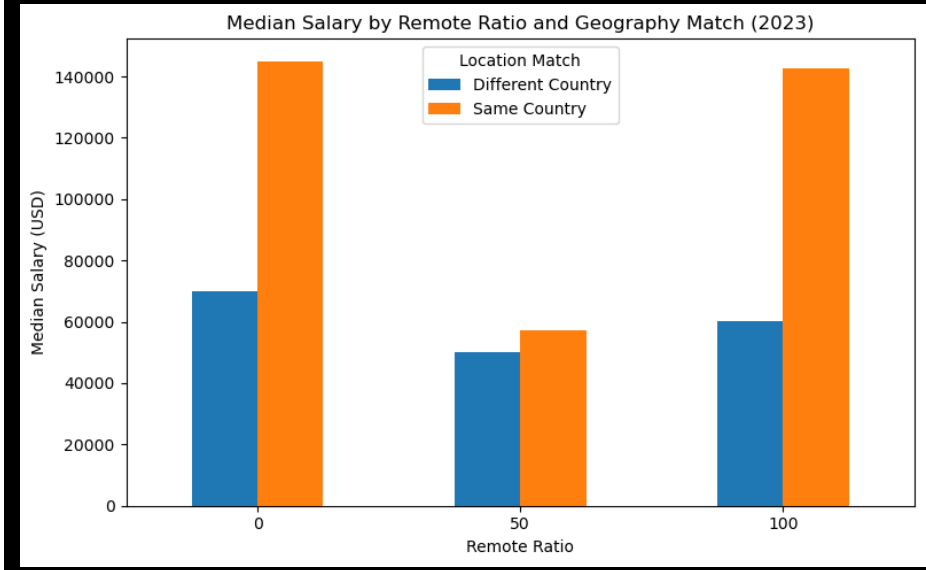
- **Large companies remain competitive but not always the top-paying**
- **Wider salary dispersion in large and medium companies.**
- **Company size matters, but its effect is not uniform**



Takeaway: Medium and large firms tend to offer higher pay, while small firms generally pay less.

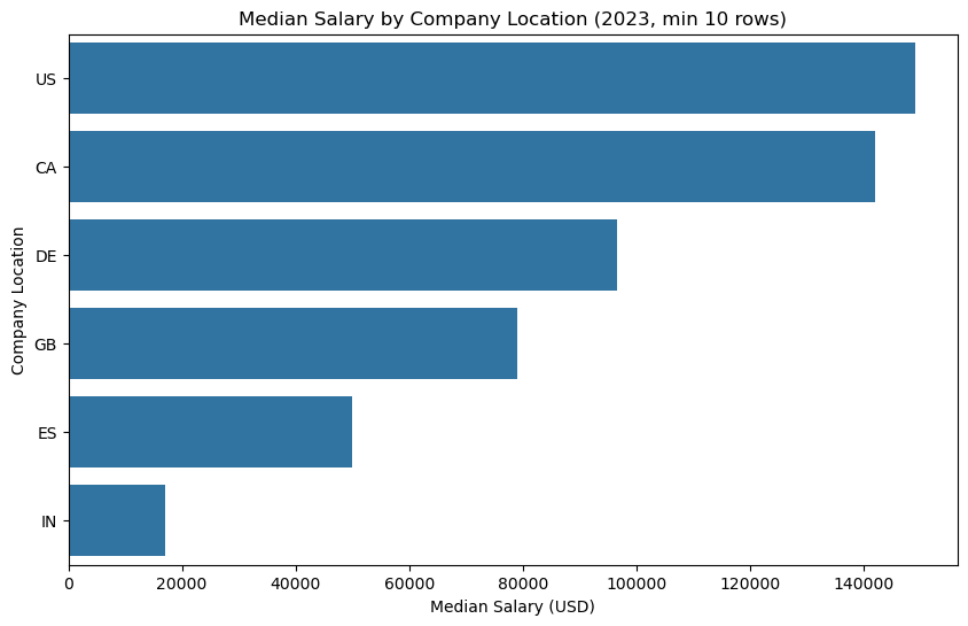
Question 5: Remote vs. Country of the Company

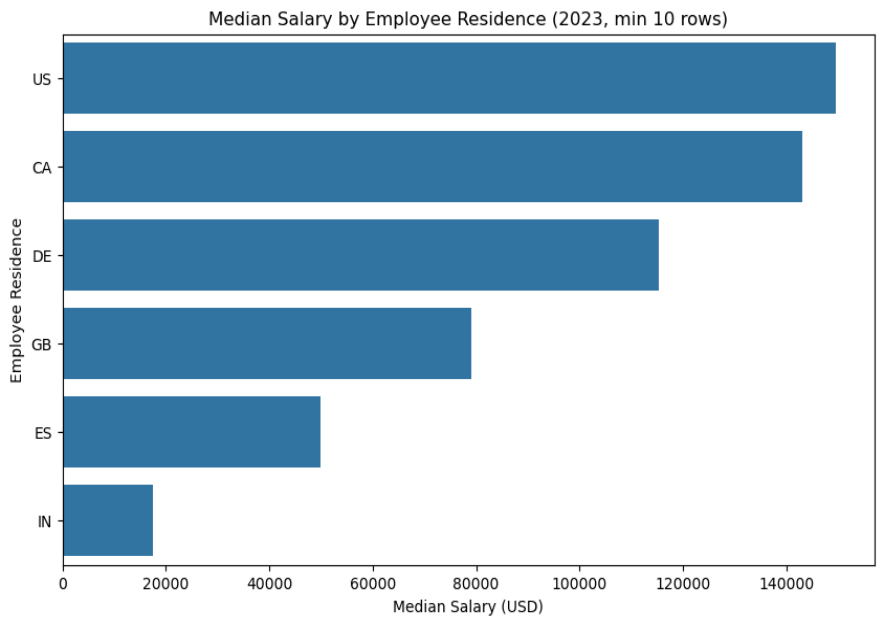




Insights:
Whether remote or not, we can see a pattern that salary for employee is higher if the employer is in the same country.

Question 5: Is pay more associated with company location or employee residence (geo pay alignment)?

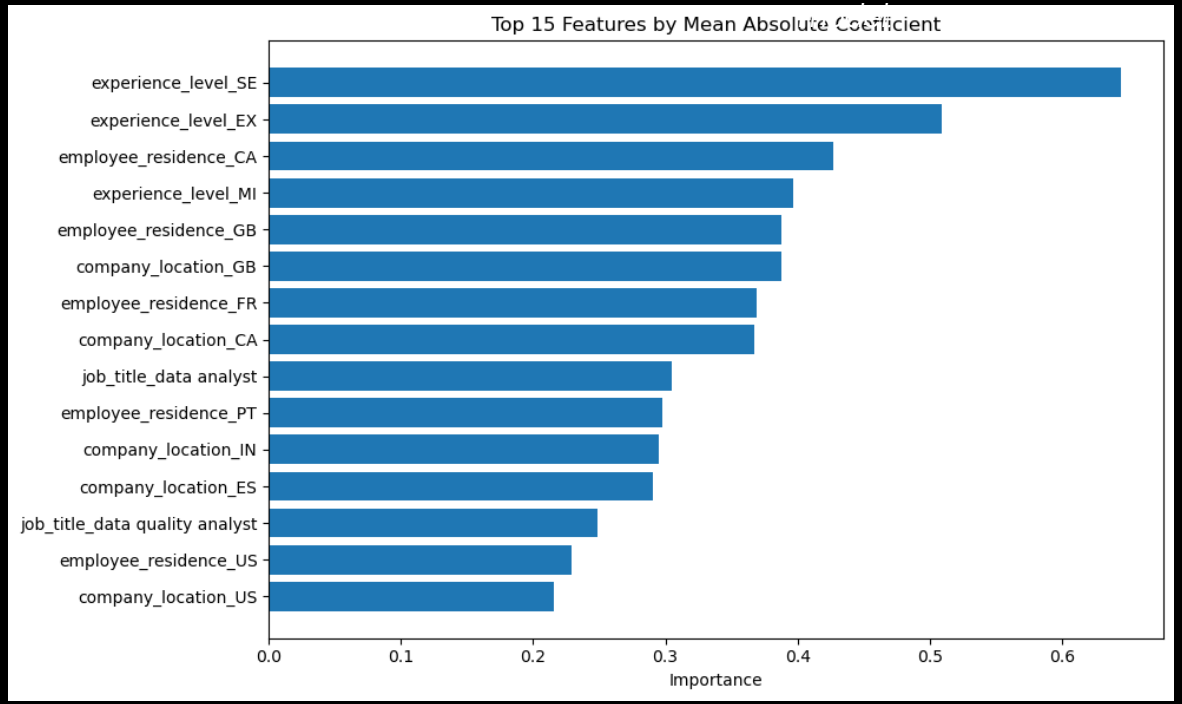




The geography-related pattern is not whether salary follows company location more than employee residence, but whether the employee and company are located in the same country.

Log Regression Model for Salary Classification

Note: *Categorical data were converted to boolean values for the*



Contributing Factors:

- Experience level
- Company location
- Residence
- Job title

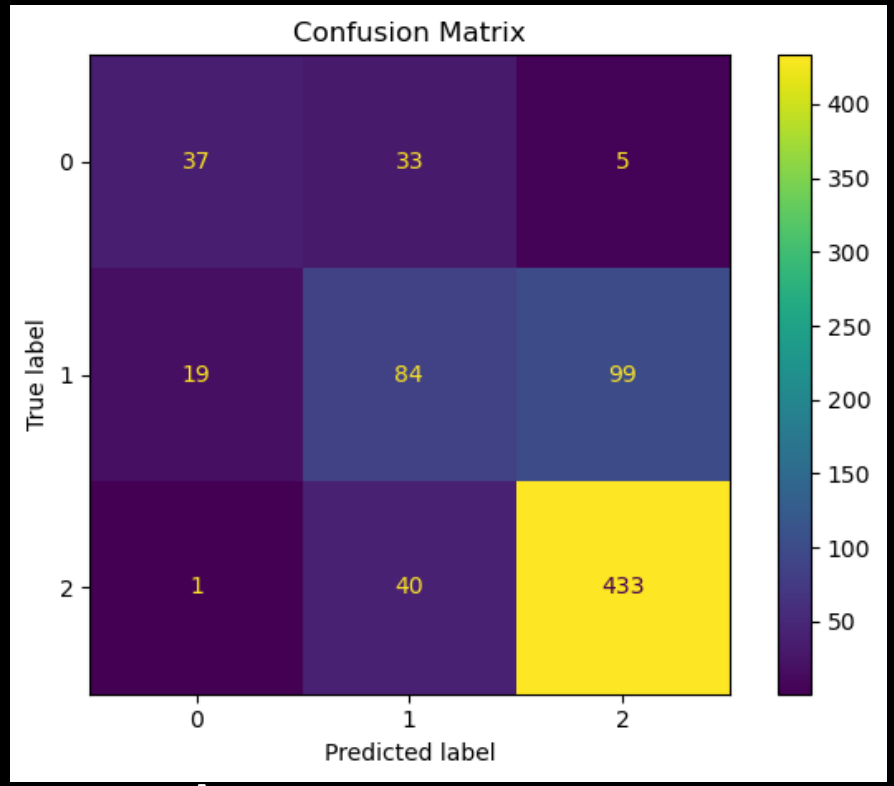
Model Predicts:

- Salary level (0~2)

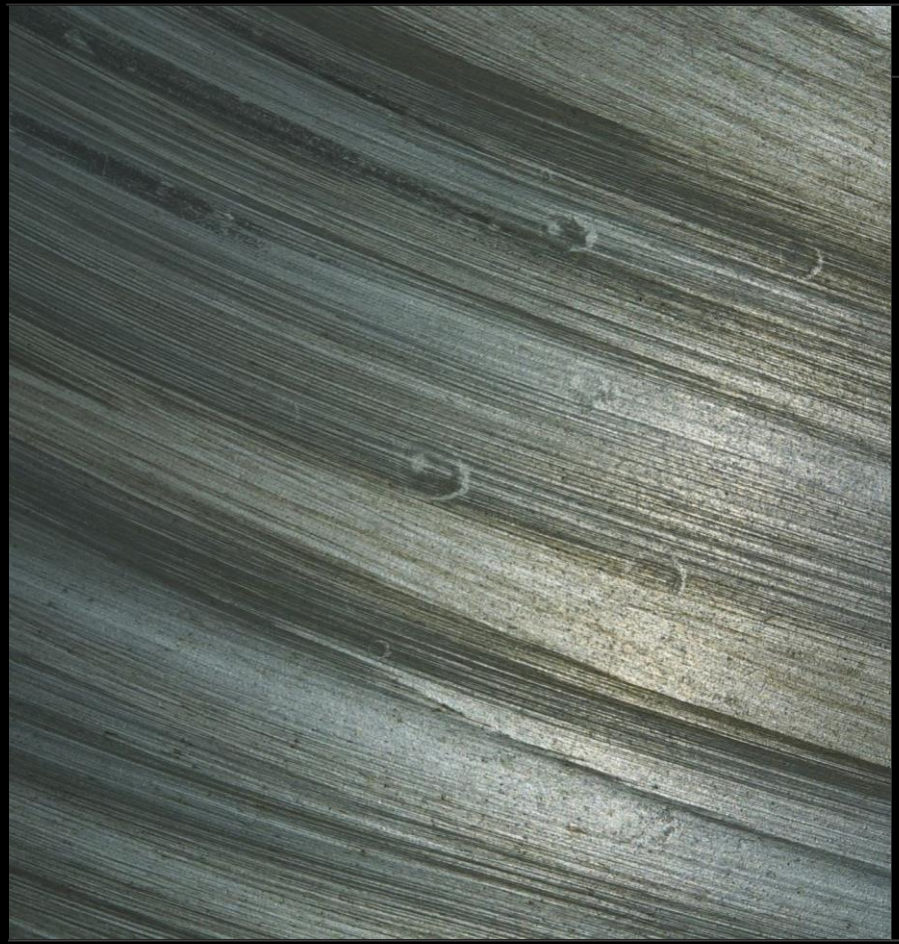
Log Regression Model for Salary Classification

Objective connection

- Our earlier analysis explains how salary varies across factors
- This model extends the project from descriptive analysis to predictive analysis
- It shows that the same factors identified in EDA can also help predict salary outcomes with reasonable accuracy.



Accuracy: ~72%



Conclusion (rank by importance of factors)

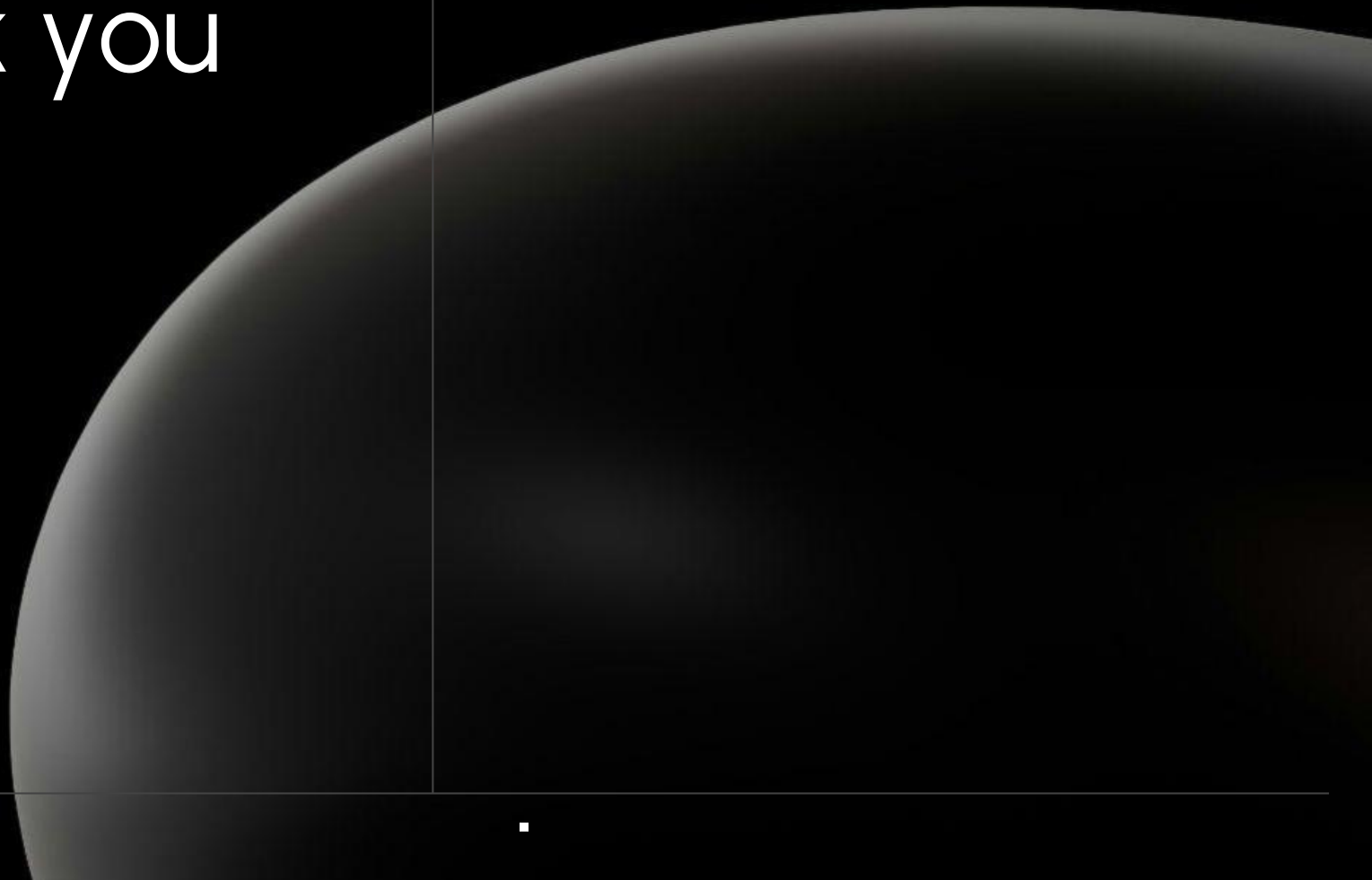
1. ***Experience level*** is the strongest predictor.

1. Senior-level and executive-level indicators have the largest coefficients

1. ***Geographic variables***, such as employee residence and company location, are also highly influential.

1. Some ***job titles*** contribute meaningful predictive power.

Thank you



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